

Sydney Technical High School



TRIAL HIGHER SCHOOL CERTIFICATE

2006

MATHEMATICS EXTENSION 2

General Instructions

- Reading time - 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown in every question
- A table of standard integrals is supplies at the back of this paper
- Start each question on a new page
- Attempt all Questions 1 – 8
- All questions are of equal value
- **Total marks 120**

Name: _____

Class: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	TOTAL

Question 1**Marks**

a) Find:

(i) $\int \frac{x \, dx}{(1+x^2)^2}$ 2

(ii) $\int \sin^3 x \, dx$ 2

(iii) $\int x\sqrt{1-x} \, dx$ 3

b) (i) Find real numbers a and b such that

$$\frac{5-3x}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$$
2

(ii) Hence find $\int \frac{5-3x}{(x+1)(x^2+1)} \, dx$ 2

c) Evaluate $\int_0^{\pi} \frac{dx}{\cos \frac{x}{2} + \sin \frac{x}{2} + 3}$ using the substitution $t = \tan\left(\frac{x}{4}\right)$ 4

Question 2

a) (i) Express $w = -1 - i$ in modulus – argument form. 2

(ii) Hence express w^{12} in the form $x + iy$ where x and y are real numbers. 2

b) Find the equation, in Cartesian form, of the locus of the point z if 2

$$|z - i| = |z + 3|.$$

c) Sketch the region in the Argand diagram that satisfies the inequality 3

$$\operatorname{Re}\left(\frac{1}{z}\right) \leq \frac{1}{2}$$

d) (i) On the Argand diagram draw a neat sketch of the locus specified by 1

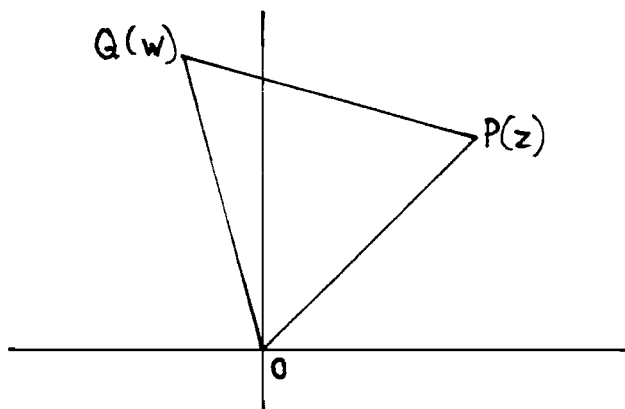
$$\arg(z + 1) = \frac{\pi}{3}$$

(ii) Hence find z so that $|z|$ is a minimum. 2

e) Points P and Q represent the complex numbers z and w respectively in the Argand Diagram. If $\triangle OPQ$ (where O is the origin) is an equilateral triangle

(i) Show why $wz = z^2 \operatorname{cis} \frac{\pi}{3}$ 1

(ii) Prove that $z^2 + w^2 = zw$ 2



Question 3

a) The hyperbola, H, has a Cartesian equation $\frac{x^2}{25} - \frac{y^2}{16} = 1$

(i) Find the coordinates of the foci S and S' 1

(ii) Show that any point, P, on H can be represented by the coordinates 3
($5 \sec \theta$, $4 \tan \theta$) and hence, or otherwise, prove that $PS - PS'$ is a constant.

(iii) Show that the equation of the normal at the point P on the hyperbola is 3

$$\frac{5x}{\sec \theta} + \frac{4y}{\tan \theta} = 41$$

(iv) If this normal meets the x axis at M and the y axis at N, prove that 3

$$\frac{PM}{PN} = \frac{16}{25}$$

b) Consider the function $y = \cos^{-1}(\cos x)$. Given the domain and range are

D: all real x

R: $0 \leq y \leq \pi$

(i) State whether the function is even, odd or neither and find its period. 2

(ii) Hence sketch the graph of the function over $-4\pi \leq x \leq 4\pi$ 1

c) Solve for x: 2

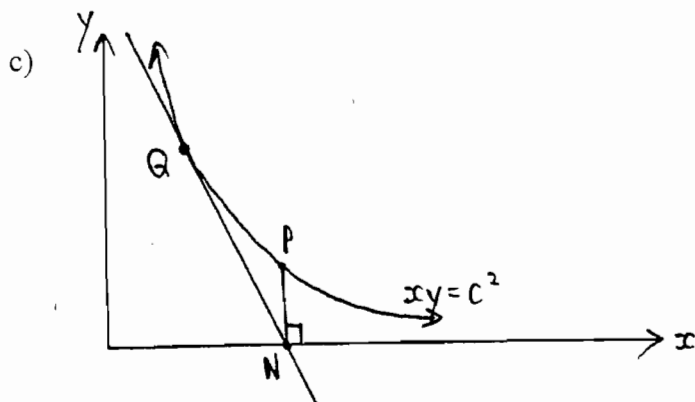
$$\tan^{-1}(3x) - \tan^{-1}(2x) = \tan^{-1}\left(\frac{1}{5}\right)$$

Question 4

- a) Find Q which is rational where 2

$$\sqrt{Q} = \sqrt{2 + \sqrt{3}} + \sqrt{2 - \sqrt{3}}$$

- b) If $f(x) = f(x-1) + x^2$ and $f(3) = 7$, evaluate $f(1)$. 2

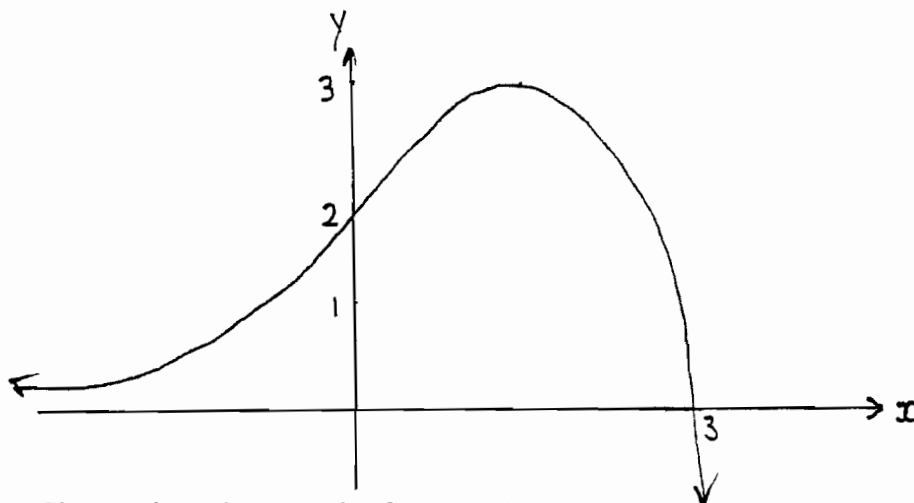


In the diagram above, $P (ct_1, \frac{c}{t_1})$ and $Q (ct_2, \frac{c}{t_2})$ are distinct variable points on the rectangular hyperbola $xy = c^2$. PN is the perpendicular from P to the x axis and the tangent at Q passes through N .

- (i) Show that $t_1 = 2t_2$ 3
- (ii) Find the Cartesian equation of the locus of T , the point of intersection of the tangents at P and Q . 3
- d) (i) By solving the equation $z^3 = 1$, find the 3 cube roots of 1. 2
- (ii) Let w be a cube root of 1 where w is not real. 1
Show that $1 + w + w^2 = 0$
- (iii) Find the quadratic equation, with integer coefficients, that has roots $4 + w$ and $4 + w^2$ 2

Question 5

a)



Shown above is a sketch of $y = f(x)$.

On separate diagrams draw sketches of:

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y = [f(x)]^3$ 2

(iii) $y = f(|x|)$ 2

(iv) $y = \log_e [f(x)]$ 2

- b) The deck of a ship was 3m below the level of a wharf at low tide and 1m above the wharf level at high tide. Low tide was at 9:30am and high tide at 4:00pm. Find the first time after low tide when the deck was level with the wharf, if the motion of the tide was simple harmonic. 4

- c) Prove by mathematical induction that, for all integers $n \geq 1$, 3

$$(\cos \theta - i \sin \theta)^n = \cos(n\theta) - i \sin(n\theta)$$

Question 6

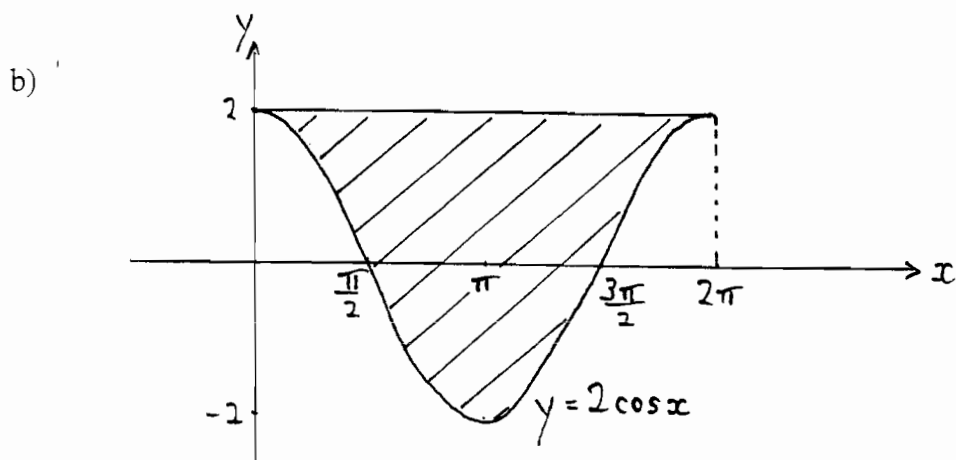
- a) Find the integers m and n such that $(x+1)^2$ is a factor of $x^5 + 2x^2 + mx + n$ 2
- b) None of the roots α , β and γ of the equation $x^3 + 3px + q = 0$ is zero.
- (i) Obtain the monic equation whose roots are $\frac{\beta\gamma}{\alpha}$, $\frac{\alpha\gamma}{\beta}$ and $\frac{\alpha\beta}{\gamma}$ 4
expressing its coefficients in terms of p and q .
- (ii) Show that if $\gamma = \alpha\beta$ then $(3p - q)^2 + q = 0$. 2
- c) For the equation $x^3 - 6x^2 + 9x - 5 = 0$
- (i) By considering stationary points, show that the equation 3
has only one real root α .
- (ii) Determine the two consecutive integers between which α lies. 1
- (iii) By considering the product of the roots of the equation, 3
express the modulus of each of the complex roots in terms of α and
deduce that the value of this modulus lies between 1 and $\frac{\sqrt{5}}{2}$.

Question 7

- a) (i) Let $I_n = \int_1^e x(\ln x)^n dx$, $n = 0, 1, 2, 3 \dots$ 2

Use integration by parts to show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$, $n = 1, 2, 3 \dots$

- (ii) The area bounded by the curve $y = \sqrt{x}(\ln x)^2$, the x axis and the lines $x=1$ and $x=e$ is rotated about the x axis. 3
Find the exact value of the volume of the solid of revolution so formed.



The shaded region is rotated about the y axis to obtain a solid of revolution.

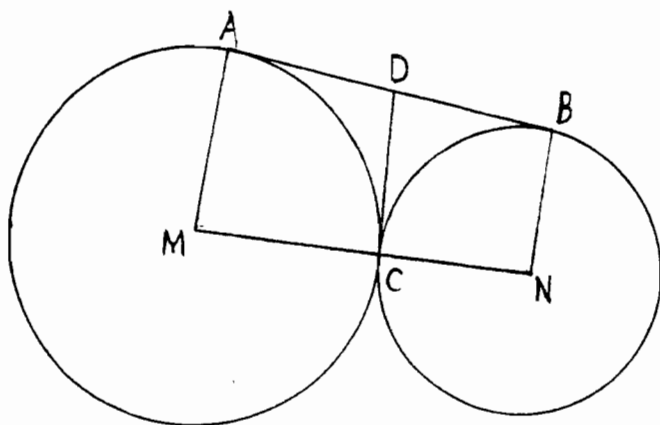
- (i) Use the method of cylindrical shells to show that the volume of this solid is given by 2

$$4\pi \int_0^{2\pi} x(1 - \cos x) dx.$$

- (ii) Hence calculate this volume. 2

Question 7 (cont.)

c)



In the diagram MCN is a straight line. Circles are drawn with centre M, radius MC and centre N, radius NC. AB is a common tangent to the two circles with points of contact at A and B respectively. CD is the common tangent at C, and meets AB at D.

- | | | |
|-------|--|---|
| (i) | Explain why AMCD and BNCD are cyclic quadrilaterals. | 2 |
| (ii) | Show that $\triangle ACD \parallel \triangle CBN$ | 2 |
| (iii) | Show that $MD \parallel CB$ | 2 |

Question 8

A particle of mass m is projected vertically upwards under gravity. The air resistance to the motion is $\frac{1}{100} mgv^2$ where v is the speed of the particle.

- (a) (i) Show that during the upward motion of the particle, if x is the upward vertical displacement of the particle from its projection point at time t , then 1

$$\ddot{x} = \frac{-1}{100} g(100 + v^2)$$

- (ii) If the initial speed of projection is u , show that the greatest height (above the projection point) reached by the particle is 5

$$\frac{50}{g} \ln \left(\frac{100 + u^2}{100} \right).$$

- (iii) Show that during the downward motion of the particle, if x is the downward vertical displacement of the particle from its highest position at a time t after it begins the downward motion, then 1

$$\ddot{x} = \frac{1}{100} g(100 - v^2)$$

- (iv) Show that the speed of the particle on return to its point of projection is 5

$$\frac{10u}{\sqrt{100 + u^2}}$$

- (v) Find the terminal velocity V of the particle for the downward motion. 1

- (vi) If the initial speed of projection of the particle is V , as found in part (v), 2
show that the speed on return to the point of projection is $\frac{1}{\sqrt{2}}V$.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

STHS 2006 Ext. 2 Trial Solutions

Question 1(a. i) $\int \frac{x dx}{(1+x^2)^2}$

$$= \frac{1}{2} \int (1+x^2)^{-2} 2x dx \quad (1)$$

$$= \frac{1}{2} \frac{(1+x^2)^{-1}}{-1} + C$$

$$= -\frac{1}{2(1+x^2)} + C \quad (1)$$

(ii) $\int \sin^3 x dx$

$$\int \sin x (1 - \cos^2 x) dx \quad (1)$$

$$= \int \sin x dx + \int \cos^2 x (-\sin x) dx$$

$$= -\cos x + \frac{1}{3} \cos^3 x + C \quad (1)$$

(iii) $\int x \sqrt{1-x} dx$

$$u = 1-x$$

$$du = -dx$$

$$= \int (1-u) \sqrt{u} du$$

$$(1)$$

$$\int u^{\frac{1}{2}} - u^{\frac{3}{2}} du \quad (1)$$

$$= \frac{2}{3} u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}} + C$$

$$= \frac{2}{3} (1-x)^{\frac{3}{2}} - \frac{2}{5} (1-x)^{\frac{5}{2}} + C \quad (1)$$

b. (i) $\frac{5-3x}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$

$$5-3x = a(x^2+1) + (bx+c)(x+1)$$

$$= ax^2 + a + bx^2 + bx + cx + c$$

$$= (a+b)x^2 + (b+c)x + (a+c)$$

$$a+b=0 \Rightarrow b=-a$$

$$b+c=-3 \Rightarrow -a+c=-3$$

$$a+c=5$$

$$2c=2$$

$$c=1$$

$$b=-4 \quad a=4$$

$$(1)$$

$$(1)$$

(ii) $\int \frac{4}{x+1} - \frac{4x}{x^2+1} + \frac{1}{x^2+1} dx$

$$= 4 \ln|x+1| - 2 \ln|x^2+1| + \tan^{-1} x + C \quad (1)$$

$$= 2 \ln \left| \frac{(x+1)^2}{x^2+1} \right| + \tan^{-1} x + C \quad (1)$$

d) $\int_0^{\pi} \frac{dx}{\cos \frac{x}{2} + \sin \frac{x}{2} + 3} = \int_0^1 \frac{4dt}{\frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2} + 3}$

$$= \int_0^1 \frac{4dt}{1-t^2+2t+3+3t^2} \quad (1)$$

$$t = \tan \frac{x}{4} \quad (1)$$

$$dt = \frac{1}{4} \sec^2 \frac{x}{4} dx$$

$$4dt = (1 + \tan^2 \frac{x}{4}) dx$$

$$dx = \frac{4dt}{1+t^2}$$

$$\int_0^1 \frac{2dt}{t^2+t+2}$$

$$= \int_0^1 \frac{2dt}{(t+\frac{1}{2})^2 + \frac{7}{4}} \quad (1)$$

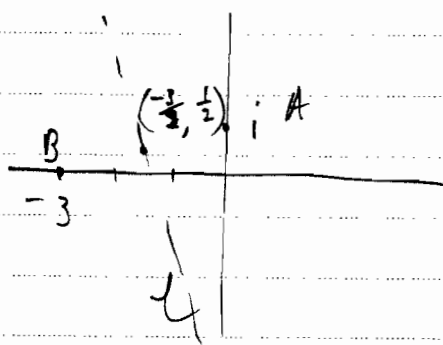
$$= \frac{4}{\sqrt{7}} \left[\tan^{-1} \frac{2t+1}{\sqrt{7}} \right]_0^1$$

$$= \frac{4}{\sqrt{7}} \left(\tan^{-1} \frac{3}{\sqrt{7}} - \tan^{-1} \frac{1}{\sqrt{7}} \right) \quad (1)$$

Question 2. a) i) $w = -1 - i$
 $= \sqrt{2} \operatorname{cis} \left(-\frac{3\pi}{4} \right) \quad (2)$

ii) $w^{12} = \left(\sqrt{2} \operatorname{cis} \left(-\frac{3\pi}{4} \right) \right)^{12}$
 $= 2^6 \operatorname{cis} -\frac{36\pi}{4}$
 $= 64 \operatorname{cis} (-9\pi)$
 $= 64 \operatorname{cis} \pi$
 $= -64 \quad (2)$

b) $|z-i| = |z+3|$



$$m_{AB} = \frac{1}{3}$$

$$m_{\perp} = -3$$

$$\therefore y - \frac{1}{2} = -3 \left(x + \frac{3}{2} \right)$$

$$y - \frac{1}{2} = -3x - \frac{9}{2}$$

$$y = -3x - 4 \quad (2)$$

c) $\operatorname{Re} \left(\frac{1}{z} \right) \leq \frac{1}{2}$

$$\operatorname{Re} \left(\frac{1}{x+iy} \right) \leq \frac{1}{2}$$

$$\operatorname{Re} \left(\frac{1}{x+iy} \cdot \frac{x-iy}{x-iy} \right) \leq \frac{1}{2}$$

$$\operatorname{Re} \left(\frac{x-iy}{x^2+y^2} \right) \leq \frac{1}{2} \quad (1)$$

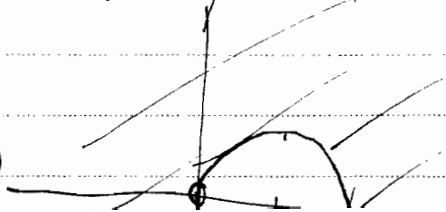
$$\frac{x}{x^2+y^2} \leq \frac{1}{2}$$

$$2x \leq x^2+y^2$$

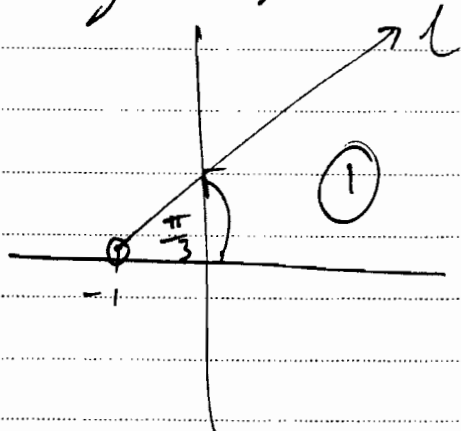
$$x^2-2x+1+y^2 \geq 1$$

(1) $(x-1)^2 + y^2 \geq 1 \quad (z \neq 0)$

(1)



i) $\arg(z+1) = \frac{\pi}{3}$



ii) Equation of l is

$$y - 0 = \tan \frac{\pi}{3} (x - -1)$$

$$y = \sqrt{3} (x+1)$$

$|z|$ is a minimum at A
where $OA \perp l$
 A is $(a, \sqrt{3}(a+1))$

$$m_{OA} = \frac{\sqrt{3}(a+1)}{a} = -\frac{1}{\sqrt{3}} \quad (1)$$

$$\therefore 3(a+1) = -a$$

$$4a + 3 = 0$$

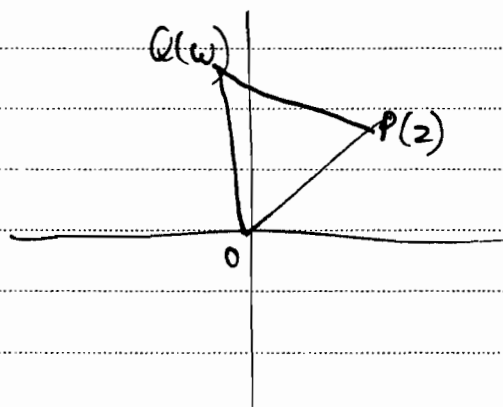
$$a = -\frac{3}{4}$$

$$\therefore A \text{ is } \left(-\frac{3}{4}, \frac{\sqrt{3}}{4}\right)$$

$$\therefore z \text{ is } -\frac{3}{4} + \frac{\sqrt{3}}{4}i \quad (1)$$

c) $w = z \operatorname{cis} \frac{\pi}{3}$

as $|w| = |z|$ and
 $\angle QOP = \frac{\pi}{3}$ (equilateral)



$$\therefore wz = (z \operatorname{cis} \frac{\pi}{3}) z$$

$$= z^2 \operatorname{cis} \frac{\pi}{3} \quad (1)$$

ii) $z^2 + w^2 = z^2 + z^2 \operatorname{cis} \frac{2\pi}{3}$

$$= z^2 (1 + \operatorname{cis} \frac{2\pi}{3})$$

$$= z^2 (1 + \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$$

$$= z^2 (1 - \frac{1}{2} + i \frac{\sqrt{3}}{2})$$

$$= z^2 (\frac{1}{2} + i \frac{\sqrt{3}}{2}) \quad (1)$$

$$= z^2 \operatorname{cis} \frac{\pi}{3}$$

$$= wz \text{ from part (i)} \quad (1)$$

Question 3. a.i) $a^2 = 25$ $b^2 = 16$ cii) Sub $(5 \sec \theta, 4 \tan \theta)$

$$b^2 = a^2(e^2 - 1)$$

$$16 = 25(e^2 - 1)$$

$$\frac{16}{25} = e^2 - 1$$

$$e^2 = \frac{41}{25}$$

$$e = \frac{\sqrt{41}}{5}$$

$$\therefore \text{Foci} = (\pm ae, 0)$$

$$\text{are } (\pm \sqrt{41}, 0) \quad (1)$$

into H

$$\frac{25 \sec^2 \theta}{25} - \frac{16 \tan^2 \theta}{16} = 1$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

is true $\forall \theta \therefore$

$P(5 \sec \theta, 4 \tan \theta)$ always satisfies H. (1)

cii) cont'd.

From the def'n of

a Hyperbola,

$$|PS - PS'|$$

$$= e P_{\text{directrix}_1} - e P_{\text{directrix}_2} \quad (1)$$

$$= e \left(5 \sec \theta - \frac{5}{\frac{\sqrt{41}}{5}} \right) - e \left(5 \sec \theta + \frac{5}{\frac{\sqrt{41}}{5}} \right)$$

$$= \frac{\sqrt{41}}{5} \times 5 \sec \theta - \frac{\sqrt{41}}{5} \times \frac{5}{\frac{\sqrt{41}}{5}} - \frac{\sqrt{41}}{5} \times 5 \sec \theta - \frac{\sqrt{41}}{5} \times \frac{5}{\frac{\sqrt{41}}{5}}$$

$$= 10 \therefore \text{a constant.} \quad (1)$$

ciii) Differentiating implicitly,

$$\frac{2x}{25} - 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{+2x}{25} \times \frac{16}{2y} = \frac{16x}{25y} \quad (1)$$

At P, the tangent

$$= \frac{80 \sec \theta}{100 \tan \theta}$$

$$= \frac{4}{5} \times \frac{1}{\cos \theta} \times \frac{\cos \theta}{\sin \theta}$$

$$= \frac{4}{5 \sin \theta}$$

$$m_{\text{normal}} = -\frac{5 \sin \theta}{4} \quad (1)$$

Equation of normal is

$$y - 4 \tan \theta = -\frac{5 \sin \theta}{4} (x - 5 \sec \theta)$$

$$4y - 16 \tan \theta = -5 \sin \theta x + 25 \tan \theta$$

$$5 \sin \theta x + 4y = 41 \tan \theta$$

$$\frac{5 \sin \theta x}{\frac{\sin \theta}{\cos \theta}} + \frac{4y}{\tan \theta} = 41$$

$$5 \cos \theta x + \frac{4y}{\tan \theta} = 41$$

$$\frac{5x}{\sec \theta} + \frac{4y}{\tan \theta} = 41. \quad (1)$$

civ) M is $(\frac{41 \sec \theta}{5}, 0)$

N is $(0, \frac{41 \tan \theta}{4}) \quad (1)$

civ) cont'd.

using $x = \frac{kx_2 + lx_1}{k+l}$

$$5 \sec \theta = \frac{k \times 0 + l \times \frac{41 \sec \theta}{5}}{k+l}$$

$$5 \sec \theta k + 5 \sec \theta l = \frac{41 \sec \theta}{5} l$$

$$5 \sec \theta k = \frac{16 \sec \theta}{5} l$$

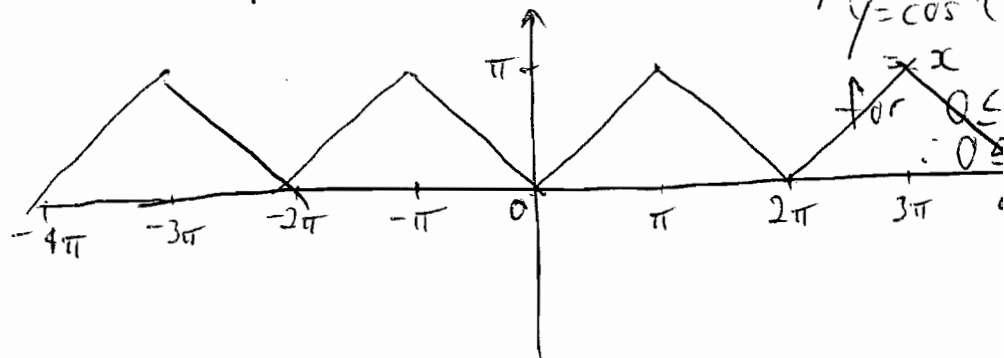
$$\frac{k}{l} = \frac{16}{25}$$

$$\therefore \frac{PM}{PN} = \frac{16}{25} \checkmark$$

b) i) Period of $y = \cos^{-1}(\cos x)$ is same as period of $\cos x$ i.e. 2π

$y = \cos^{-1}(\cos x)$ is even as $\cos$ is even. i.e. $\cos^{-1}(\cos x) = \cos^{-1}(\cos(-x)) = \cos^{-1}(\cos x)$

ii) Symmetrical across the y axis.



$$3c. \tan^{-1} 3x - \tan^{-1} 2x = \tan^{-1} \frac{1}{5}$$

$$\tan[\tan^{-1} 3x - \tan^{-1} 2x] = \tan[\tan^{-1} \frac{1}{5}]$$

$$\frac{\tan[\tan^{-1} 3x] - \tan[\tan^{-1} 2x]}{1 + \tan[\tan^{-1} 3x] \tan[\tan^{-1} 2x]} = \frac{1}{5}$$

$$\frac{3x - 2x}{1 + 3x \cdot 2x} = \frac{1}{5} \quad (1)$$

$$5x = 1 + 6x^2$$

$$6x^2 - 5x + 1 = 0$$

$$(3x-1)(2x-1) = 0$$

$$x = \frac{1}{3} \text{ or } \frac{1}{2} \quad (1)$$

Question 4. a) $\sqrt{Q} = \sqrt{2+\sqrt{3}} + \sqrt{2-\sqrt{3}}$

$$Q = 2 + \sqrt{3} + (2 - \sqrt{3}) + 2\sqrt{2+\sqrt{3}}\sqrt{2-\sqrt{3}} \quad (1)$$

$$= 4 + 2\sqrt{4-3}$$

$$= 4 + 2 = \underline{6} \quad (1)$$

$$b) f(x) = f(x-1) + x^2$$

$$f(3) = f(2) + 3^2 = 7$$

$$\therefore f(2) = -2 \quad (1)$$

$$f(2) = f(2-1) + 2^2$$

$$-2 = f(1) + 4$$

$$\therefore f(1) = -6 \quad (1)$$

c. (i) $x = ct$ $y = ct^{-1}$
 $\frac{dx}{dt} = c$ $\frac{dy}{dt} = -\frac{c}{t^2}$

$$\frac{dy}{dx} = -\frac{c}{t^2} \times \frac{1}{c}$$

$$= -\frac{1}{t^2} \quad (1)$$

tangent:

$$y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$$

$$t^2y - ct = -x + ct$$

$$x + t^2y = 2ct \quad (1)$$

tangent at Q:

$$x + t_2^2y = 2ct_2$$

N $(ct, 0)$:

$$ct_1 + 0 = 2ct_2$$

$$t_1 = 2t_2 \quad (1)$$

(ii) tangent at Q

$$x + t_2^2y = 2ct_2 \quad (1)$$

tangent at P

$$x + t_1^2y = 2ct_1$$

ie. $x + 4t_2^2y = 4ct_2 \quad (2)$

since $t_1 = 2t_2$ (1) mark

$$(2) - (1)$$

$$3t_2^2y = 2ct_2$$

$$y = \frac{2c}{3t_2}$$

sub. in (1)

$$x + \frac{2ct_2}{3} = 2ct_2$$

$$x = \frac{4ct_2}{3} \quad (1) \text{ mark}$$

$$xy = \frac{2c}{3t_2} \cdot \frac{4ct_2}{3}$$

$$xy = \frac{8c^2}{9} \quad (1) \text{ mark}$$

d. (i) $z^3 - 1 = 0$

$$(z-1)(z^2+z+1) = 0 \quad (1)$$

$$z = 1 \text{ or } \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= 1 \text{ or } \frac{-1 \pm i\sqrt{3}}{2} \quad (1)$$

(ii) $(\omega-1)(\omega^2+\omega+1) = 0$ from (i)

Now $\omega \neq 1 \therefore$

$$\omega^2 + \omega + 1 = 0 \quad (1)$$

(iii) $\alpha + \beta = 4 + \omega + 4 + \omega^2$

$$= 7 + 1 + \omega + \omega^2 \quad (1)$$

$$= 7$$

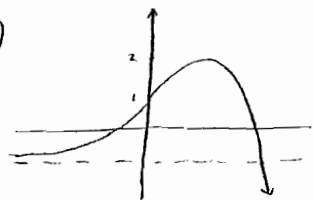
$$2\beta = 16 + 4\omega + 4\omega^2 + \omega^3$$

$$= 12 + 4(1 + \omega + \omega^2) + 1$$

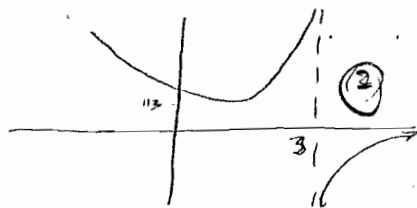
$$= 13$$

$$\therefore z^2 - 7z + 13 = 0 \quad (1)$$

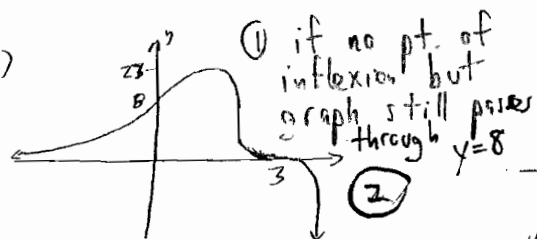
a) (i)



(1)

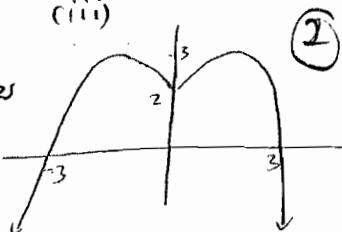


(ii)

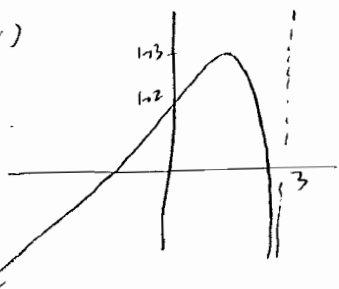


① if no pt. of inflexion but graph still passes through $y=8$

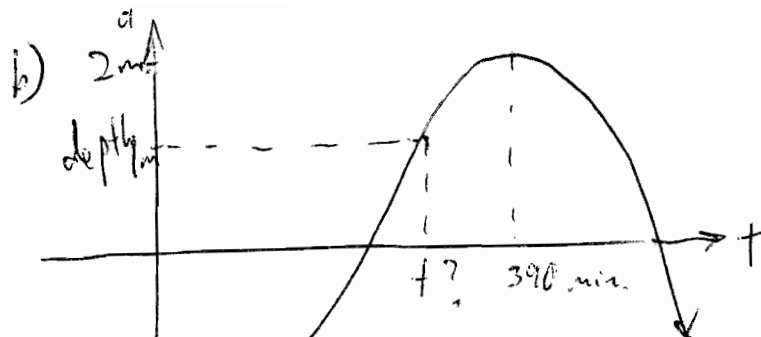
(iii)



(iv)



②



$$\text{Period} = 780 = \frac{2\pi}{n}$$

$$\therefore n = \frac{2\pi}{780} \quad ①$$

$$\therefore d = -2 \cos \frac{2\pi}{780} t \quad ①$$

when $d=1$ solve for t

$$-\frac{1}{2} = \cos \frac{2\pi}{780} t$$

$$\frac{2\pi}{780} t = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\therefore t = \frac{780}{3} = 260 \text{ minutes} \quad ①$$

ie. 4 hrs 20 minutes after

9:30 am

ie. 1:50 pm ^① is when depth is level with wharf.

QUESTION 6

(a) Let $P(x) = x^5 + 2x^2 + mx + n$; $\therefore P'(x) = 5x^4 + 4x + m$.

$(x+1)^2$ is a factor of $P(x)$; $\therefore P(-1) = 0$ and $P'(-1) = 0$. (2)

$P'(-1) = 0$, $\therefore m = -1$. $P(-1) = 0$, $\therefore n = -2$.

(b)(i) $x^3 + 3px + q = 0$ has roots α, β, γ ($\alpha \neq 0, \beta \neq 0, \gamma \neq 0$);

$\therefore \alpha + \beta + \gamma = 0$, $\alpha\beta + \alpha\gamma + \beta\gamma = 3p$, $\alpha\beta\gamma = -q$ ($q \neq 0$). (1)

$$\frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{\beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2}{\alpha\beta\gamma}$$

$$= \frac{(\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{\alpha\beta\gamma} = \frac{-9p^2}{q}. \quad (1)$$

$$\frac{\beta\gamma}{\alpha} \cdot \frac{\alpha\gamma}{\beta} + \frac{\beta\gamma}{\alpha} \cdot \frac{\alpha\beta}{\gamma} + \frac{\alpha\gamma}{\beta} \cdot \frac{\alpha\beta}{\gamma} = \gamma^2 + \beta^2 + \alpha^2$$

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) = -6p. \quad (1)$$

$$\frac{\beta\gamma}{\alpha} \cdot \frac{\alpha\gamma}{\beta} \cdot \frac{\alpha\beta}{\gamma} = \alpha\beta\gamma = -q.$$

$\therefore$ the required equation is

$$x^3 - \left(\frac{-9p^2}{q}\right)x^2 + (-6p)x - (-q) = 0, \therefore x^3 + \frac{9p^2}{q}x^2 - 6px + q = 0. \quad (1)$$

(ii) $\gamma = \alpha\beta$ if and only if $x = 1$ is a root of this equation
ie. if and only if $(3p - q)^2 + q = 0$. (2)

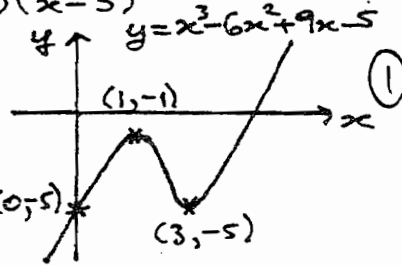
(c) (i) Let $f(x) = x^3 - 6x^2 + 9x - 5$

$$\therefore f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$$

sign of $f'(x)$ $\xrightarrow{+ve, -ve, +ve} x$

$(1, -1)$ is a maximum turning point, $(3, -5)$ is a minimum turning point.

$\therefore x^3 - 6x^2 + 9x - 5 = 0$ has only one real root α . (1)



(ii) $f(4) = -1$, $f(5) = 15$; $\therefore 4 < \alpha < 5$ (1)

(iii) Let the complex roots be $p+iq$ and $p-iq$;

$\therefore$ product of roots $(p+iq)(p-iq)\alpha = 5$

$\therefore p^2 + q^2 = \frac{5}{\alpha}$, $\therefore \sqrt{p^2 + q^2} = \frac{\sqrt{5}}{\sqrt{\alpha}}$ (1)

now $4 < \alpha < 5$, $\therefore \frac{1}{5} < \frac{1}{\alpha} < \frac{1}{4}$, $\therefore \frac{1}{\sqrt{5}} < \frac{1}{\sqrt{\alpha}} < \frac{1}{2}$, (1)

$\therefore 1 < \frac{\sqrt{5}}{\sqrt{\alpha}} < \frac{\sqrt{5}}{2}$, $\therefore 1 < \sqrt{p^2 + q^2} < \frac{\sqrt{5}}{2}$. (1)

Question 7.

(a) (i) $I_n = \int_1^e x (\ln x)^n dx$

$$= \left[\frac{x^2}{2} (\ln x)^n \right]_1^e - \int_1^e \frac{x^2}{2} n (\ln x)^{n-1} \frac{1}{x} dx \quad (1)$$

$$= \frac{e^2}{2} - \frac{n}{2} I_{n-1} \quad (1) \text{ as req'd.}$$

(ii) $V = \pi \int_1^e y^2 dx$

$$= \pi \int_1^e x (\ln x)^4 dx$$

$$= \pi I_4 \quad (1)$$

$$I_4 = \left(\frac{e^2}{2} - 2I_3 \right)$$

$$= \frac{e^2}{2} - 2 \left(\frac{e^2}{2} - \frac{3}{2} I_2 \right)$$

$$= -\frac{e^2}{2} + 3I_2 \quad (1)$$

$$= -\frac{e^2}{2} + 3 \left(\frac{e^2}{2} - I_1 \right)$$

$$= e^2 + 3I_2$$

$$= e^2 - 3I_1$$

$$= e^2 - 3 \left(\frac{e^2}{2} - \frac{1}{2} I_0 \right)$$

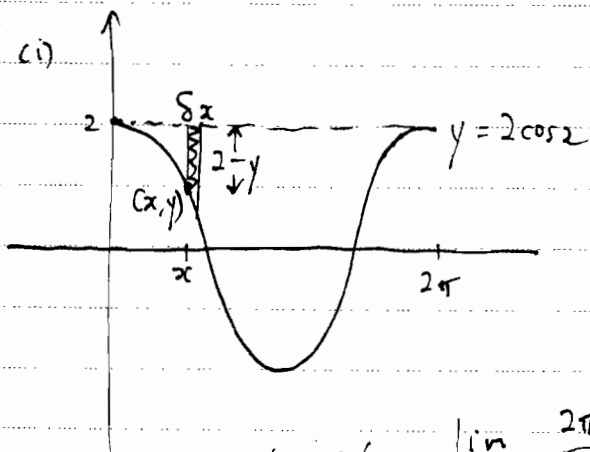
$$= -\frac{e^2}{2} + \frac{3}{2} I_0$$

$$= -\frac{e^2}{2} + \frac{3}{2} \left(\frac{e^2}{2} - \frac{1}{2} \right)$$

$$= \frac{e^2}{4} - \frac{3}{4}$$

$$\therefore \text{Volume} = \frac{\pi}{4} (e^2 - 3) \text{ units}^3 \quad (1)$$

(b) (i)



gives a shell with
Volume

$$\delta V = 2\pi x (2 - y) \delta x$$

$$= 2\pi x (2 - 2 \cos x) \delta x$$

$$= 4\pi x (1 - \cos x) \delta x \quad (1)$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{2\pi} 4\pi x (1 - \cos x) \delta x$$

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$$\therefore V = 4\pi \int_0^{2\pi} x(1 - \cos x) dx \quad (1)$$

$$(ii) = 4\pi \left\{ \left[x(x - \sin x) \right]_0^{2\pi} - \int_0^{2\pi} x - \sin x dx \right\}$$

$$= 4\pi \left\{ 2\pi \times (2\pi) - \left[\frac{x^2}{2} + \cos x \right]_0^{2\pi} \right\} \quad (1)$$

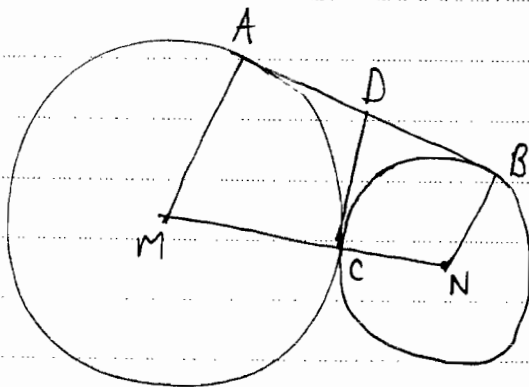
$$= 4\pi \left\{ 4\pi^2 - \left[\left(\frac{4\pi^2}{2} + 1 \right) - (1) \right] \right\}$$

$$= 4\pi \{ 4\pi^2 - 2\pi^2 \}$$

$$= 8\pi^3 \text{ units}^3$$

(1)

(C)



(i) $\angle MAD = \angle MCD = 90^\circ$ (tangent $\perp$ radius at pt. of contact)
 AMCD is cyclic (opposite angles supplementary) (1)

$\angle NBD = \angle NCD = 90^\circ$ (tangent $\perp$ radius at pt. of contact)
 BNCD is cyclic (opposite angles supplementary) (1)

(ii) $\angle CDA = \angle BNC$ ($= 2\theta$ say) (ext. angle of cyclic quad BNCD equals interior opposite angle)
 $DA = DC$ (1) (tangents drawn from external point)

$\therefore \angle DAC = \angle ACD = 90 - \theta$ (equal angles opposite equal sides, angle sum of triangle is 180°)

$NB = NC$ (equal radii of circle)

$\therefore \angle NCB = \angle CBN = 90 - \theta$ (equal angles opposite equal sides)

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ciii) $\angle NCB = \angle DAC (= 90 - \theta)$

$\angle BAC = \angle CMD$ (angles in circle through A, M, C, D standing on arc CD are equal) (1)

$\therefore \angle CMD = \angle NCB$ ($\angle CMD$ and $\angle NCB$ are a pair of equal corresponding angles on transversal MCN) (1)

$\therefore MD \parallel CB$

Question 8

a) (i) $m\ddot{x} = -mg - \frac{1}{100} mgv^2$
 $\ddot{x} = -g - \frac{1}{100} g v^2$
 $\ddot{x} = -\frac{1}{100} g (100 + v^2)$ as req'd. (1)

(ii) $v \frac{dv}{dx} = \frac{-g(100+v^2)}{100}$
 $\frac{dv}{dx} = \frac{-g(100+v^2)}{100v}$
 $\frac{dx}{dv} = \frac{-100v}{g(100+v^2)}$ (1)

$$x = \int \frac{-100v}{g(100+v^2)} dv$$

$$x = -\frac{100}{g} \frac{1}{2} \ln(100+v^2) + C$$
 (1)

when $x=0$ $v=0$

$$\therefore 0 = -\frac{50}{g} \ln(100+v^2) + C$$

$$\therefore C = \frac{50}{g} \ln(100+v^2)$$
 (1)

$$\therefore x = \frac{50}{g} \ln(100+v^2) - \frac{50}{g} \ln(100+v^2)$$

$$x = \frac{50}{g} \ln\left(\frac{100+v^2}{100+v^2}\right)$$
 (1)

Greatest height reached when $v=0$

ie:

$$x = \frac{50}{g} \ln\left(\frac{100+v^2}{100}\right) \text{ as required} \quad (1)$$

(iii) $m\ddot{x} = mg - \frac{1}{100} mgv^2$
 $\ddot{x} = g - \frac{1}{100} g v^2$

$$\ddot{x} = \frac{1}{100} g (100 - v^2) \text{ as req'd} \quad (1)$$

(iv) $v \frac{dv}{dx} = \frac{1}{100} g (100 - v^2)$

$$\frac{dv}{dx} = \frac{g}{100v} (100 - v^2) = \frac{g(100 - v^2)}{100v}$$
 (1)

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$$\therefore \frac{dx}{dv} = \frac{100v}{g(100-v^2)}$$

$$x = \frac{100}{g} \times \frac{1}{2} \log_e (100-v^2) + C \quad (1)$$

$$x = -\frac{50}{g} \ln(100-v^2) + C$$

when $x = 0, v = 0$

$$\therefore 0 = -\frac{50}{g} \ln 100 + C$$

$$\therefore C = \frac{50}{g} \ln 100 \quad (1)$$

$$x = \frac{50}{g} \ln 100 - \frac{50}{g} \ln(100-v^2)$$

$$x = \frac{50}{g} \ln \left(\frac{100}{100-v^2} \right) \quad (1)$$

Now find v when $x = \frac{50}{g} \ln \left(\frac{100+v^2}{100} \right)$
from part (ii)

$$\cancel{\frac{50}{g}} \ln \left(\frac{100+v^2}{100} \right) = \cancel{\frac{50}{g}} \ln \left(\frac{100}{100-v^2} \right)$$

$$\frac{100+v^2}{100} = \frac{100}{100-v^2}$$

$$100-v^2 = \frac{10000}{100+v^2}$$

$$v^2 = 100 - \frac{10000}{100+v^2}$$

$$v^2 = \frac{100(100+v^2) - 10000}{100+v^2}$$

$$= \frac{10000 + 100v^2 - 10000}{100+v^2}$$

$$\therefore v = \frac{10v}{\sqrt{100+v^2}} \text{ as req'd.} \quad (1)$$

(v) Terminal velocity is when $\ddot{x} = 0$. $\ddot{x}$ for downward motion is

$$\ddot{x} = \frac{1}{100} g (100 - v^2) = 0$$

$$\text{when } v = 10$$

$\therefore V = 10$ is terminal velocity (1)

(vi) If the initial speed of projection is $V = 10$, then speed upon return to the point of projection will be

$$\frac{10 \times 10}{\sqrt{100 + 10^2}} \quad \text{from (iv)} \quad (1)$$

$$= \frac{100}{\sqrt{200}}$$

$$= \frac{100}{10\sqrt{2}}$$

$$= \frac{10}{\sqrt{2}}$$

which equals $\frac{1}{\sqrt{2}} \times 10$

$$= \frac{1}{\sqrt{2}} V \quad \text{as required} \quad (1)$$